
Artificial Intelligence

Assignment 2

Total Points: 5

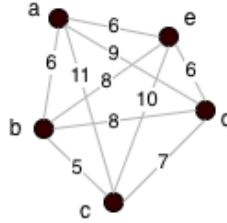
Local Search, Adversarial Search and Genetic Algorithms

Due: 17:00, 10 April, 2026

Submission PDF to: <https://f.wps.cn/g/EzPTyf4a/>

1. Local Search (1.5 pts)

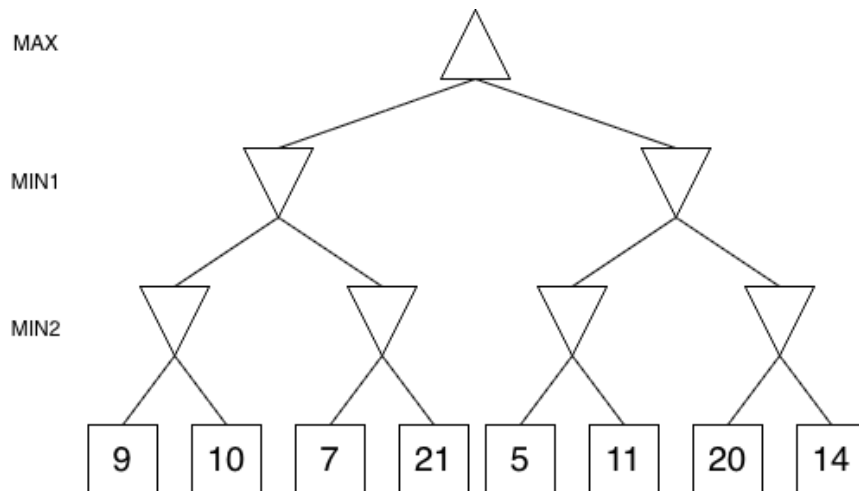
Link cables between a set of servers S to integrate all servers to a single network (each server can reach any other through a path), while keeping the lowest cost for linking cables. The cost of linking any two servers $a, b \in S$ is given by a function $c(a, b)$.



1. Formulate this problem as a local search problem: define (1) the feature representation of a candidate solution; and (2) the loss function of a candidate solution.
2. Apply greedy descent on this problem, write down the resulting candidate solution for the first three iterations.

2. Adversarial Search (1.5 pts)

Consider a game tree with one MAX node and two MIN nodes below.



1. Report the values of the non-terminal nodes.
2. If alpha-beta pruning is applied, which edges would be pruned? Place cross ("x") on the pruned edges.

3. Genetic Algorithm (2 pts)

The 0-1 knapsack problem involves a list of n items, where each item has a positive reward and a weight in kg. Suppose you hold a bag with total capacity 14 kg. Your goal is to put items from the list into your bag so that the total reward of the selected item is maximised, while respecting the capacity of the bag. Each item can only be selected for once.

Item ID	Weight	Reward(\$)
1	2	1
2	2	2
3	3	6
4	5	8
5	12	10

1. Consider this problem as an evolutionary search problem. Define the representation of chromosomes and the fitness function. For simplicity, crossover/mutation operations are not required to produce a feasible solution (≤ 14 kg).
2. Assume the population is 4. Simulate one generation of genetic algorithm in solving the above problem. You should use example chromosomes to demonstrate the selection, crossover and mutation operations.